Machine Learning Capstone Project

Project Title

Student Performance Prediction System

Project Overview

The Student Performance Prediction System is a machine learning application designed to analyze student academic data and predict performance outcomes. The system identifies factors affecting student performance and generates insights that can help improve learning outcomes.

This project gives students practical exposure to how machine learning is used in education technology and academic analytics systems.

Project Objective

Develop a machine learning application that can:

- Analyze student academic data
- Predict student performance levels
- Identify key factors affecting performance
- Generate performance insights
- Visualize student analytics through dashboards

Industry Relevance

This project is inspired by systems used in:

- EdTech platforms
- Learning management systems
- Academic analytics platforms
- Online learning applications

Students will understand how machine learning supports educational decision-making.

Project Modules

Module 1: Data Collection & Preprocessing

- Import student performance dataset
- Handle missing values
- Remove duplicate records
- Encode categorical data
- Normalize important features

Learning Outcomes

- Data cleaning
- Data preprocessing
- Dataset preparation

Module 2: Exploratory Data Analysis

Analyze factors such as:

- Attendance
- Study hours
- Internal marks
- Participation
- Previous academic performance

Create charts and graphs for insights.

Learning Outcomes

- Data visualization
- Pattern analysis
- Educational data understanding

Module 3: Performance Prediction Model

Build a machine learning model to predict:

- High performance
- Average performance
- Low performance

Suggested algorithms:

- Linear Regression
- Decision Tree
- Random Forest

Learning Outcomes

- Machine learning model building
- Model training and evaluation
- Prediction systems

Module 4: Performance Dashboard

Display:

- Predicted scores
- Student comparison
- Subject-wise analysis
- Performance trends

Learning Outcomes

- Dashboard development
- Data storytelling
- Analytics reporting

Module 5: Recommendation System

Generate suggestions such as:

- Increase study hours
- Improve attendance
- Focus on weak subjects
- Attend practice sessions

Learning Outcomes

- Insight generation
- Decision-support systems

Recommended Tech Stack

Programming

- Python

Libraries

- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- Seaborn

Dashboard Tools

- Streamlit or Power BI

Database

- MySQL (Optional)

Deployment

- Streamlit Cloud or Render

Project Workflow

1. Dataset Collection
2. Data Cleaning
3. Exploratory Data Analysis
4. Model Development
5. Prediction Generation
6. Dashboard Visualization
7. Recommendation Generation

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Dataset Suggestions

Students may use:

- Kaggle Student Performance Dataset
- Educational Performance Dataset
- Self-created academic datasets

Deliverables

Students must submit:

- Source Code
- Project Report
- PPT Presentation
- GitHub Repository
- Dashboard Screenshots
- Deployment Link
- Demo Video

Scope Limitations

Students should avoid:

- Deep Learning models
- Real-time student monitoring systems
- Large-scale educational platforms
- Complex AI architectures

Focus should remain on building a clean and functional prediction system.

25-Day Project Plan

Days	Tasks
Day 1–3	Requirement Analysis and Dataset Collection
Day 4–7	Data Cleaning and Preprocessing
Day 8–12	Exploratory Data Analysis
Day 13–17	Machine Learning Model Development
Day 18–21	Dashboard Development
Day 22–23	Recommendation System
Day 24	Testing and Bug Fixing
Day 25	Deployment and Final Presentation

Evaluation Criteria

Criteria	Weightage
Functionality	30%
Data Analysis	20%
Machine Learning Model	20%
Dashboard/UI	15%
Deployment	10%
Documentation	5%

Final Outcome

By completing this project, students will learn:

- Educational data analysis
- Machine learning model development
- Prediction system implementation
- Dashboard creation
- Insight generation
- End-to-end machine learning workflow

The final system will function as a Student Performance Prediction platform capable of analyzing academic data, predicting student outcomes, and generating performance improvement insights.